



Addressing the issue of object presence/absence in a semantically queryable map (Grade A)

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[Github - Code](#)

Overview

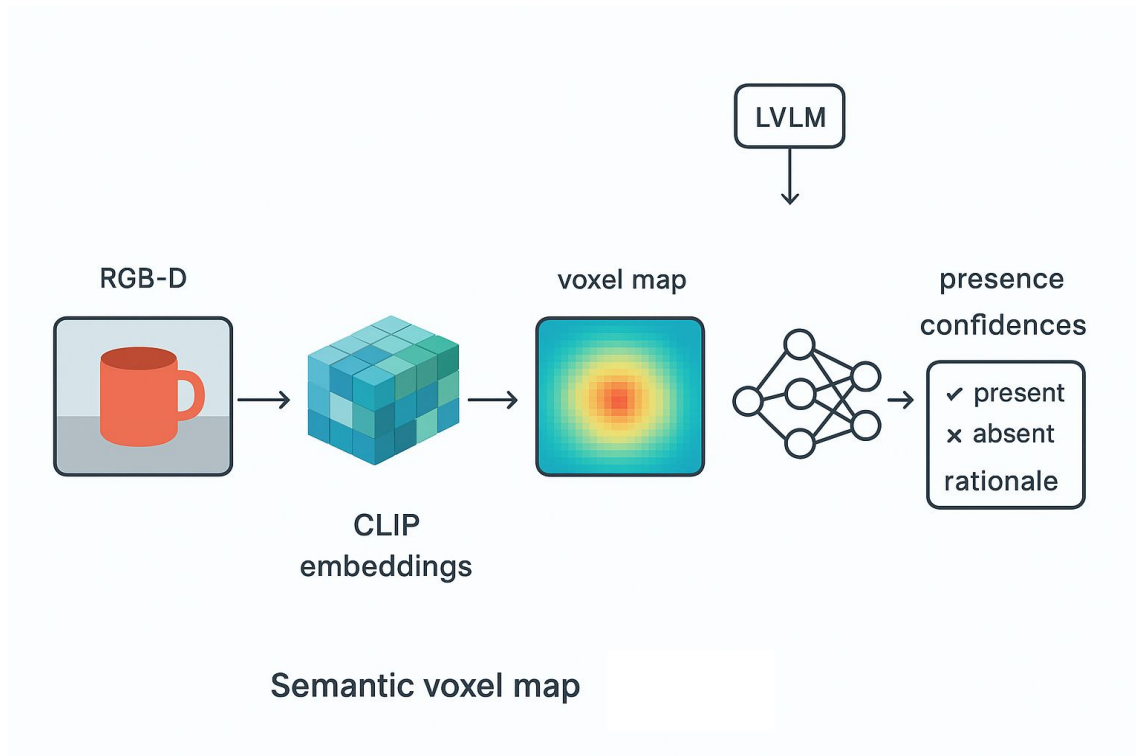
- Encoder: CLIP for text↔image embeddings to build the semantic voxel map.
- Data: 3 ARKitScenes × ~80 poses/scene × 8 queries ⇒ ~2,000 RGB + heatmap pairs.
- Class set: ball, door, floor, painting, sofa, table, wall, window.
- Labeling: presence/absence derived per (scene, query); 687 positives / 1265 negatives.
- Models: (i) VLM (Gemini) for qualitative checks (ground truth), (ii) Resnet18 trained from scratch for presence classification, (iii) threshold-based classifier

Problem Statement

Similarity maps (text $\leftrightarrow$ voxel features) are normalized scene-wise, absolute magnitudes are unreliable.

- When the object is absent, we can still get false “hot” regions due to random noise and normalization.
- We need a binary decision with calibrated confidence that separates true presence from normalization artifacts.
- Input: RGB-D frames + camera poses and a text query (class label). Output: present/absent + confidence.
- Target behaviors: reliable even in cluttered scenes, with occlusion, scale changes or ambiguous queries.

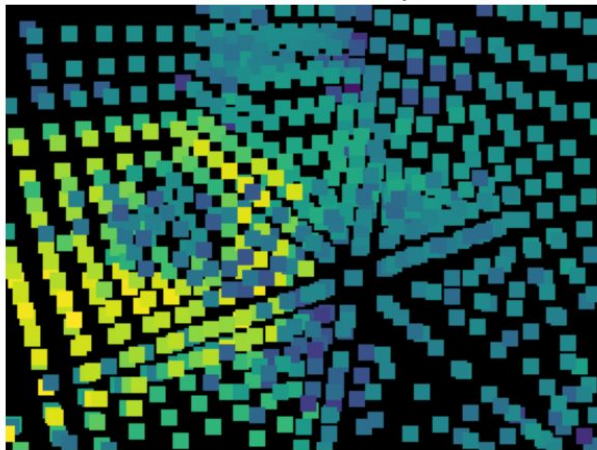
Solution Overview



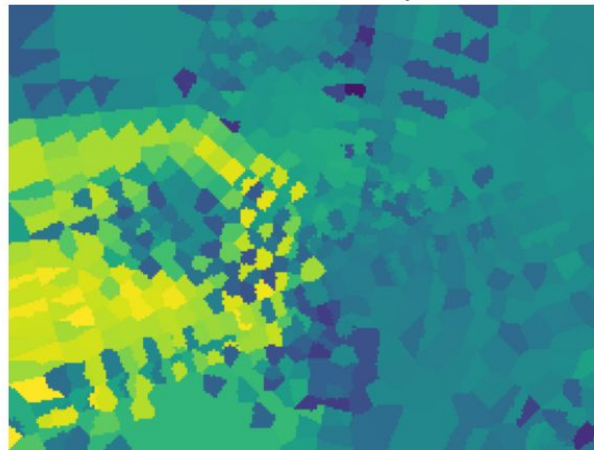
Frame rendering

- Query + Voxel map $\rightarrow$ 3D similarity map $\rightarrow$ Projected and clipped 2D similarity in the plane of the camera $\rightarrow$ 2D heatmap image (Filled black regions)

Semantic Voxel Similarity

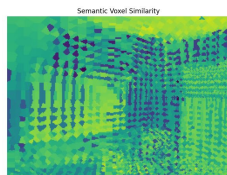


Semantic Voxel Similarity



VLM-based Classifier (Grade E)

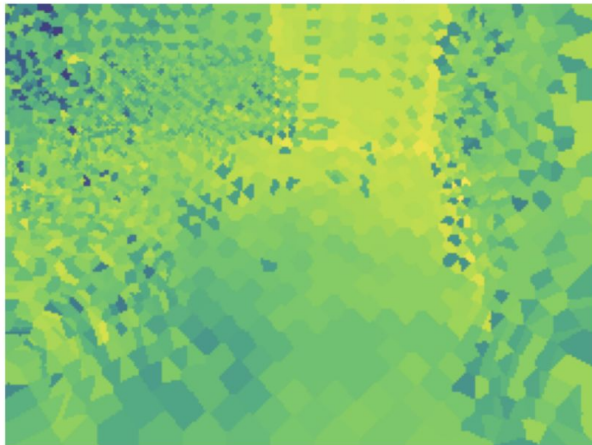
- **Inputs:**
 - RGB, rendered similarity heatmap, and class query. (eg. "table")
 - We used CLIP to compute text $\leftrightarrow$ voxel similarity; heatmaps are rendered per pose. (ext. of Lab2A)
- **VLM System prompt:**
 - "Given the RGB image and the heatmap for '<class>', is the object truly present? Does the heatmeap rightly depict the same? Answer yes/no and a short why."
- **Example output:**



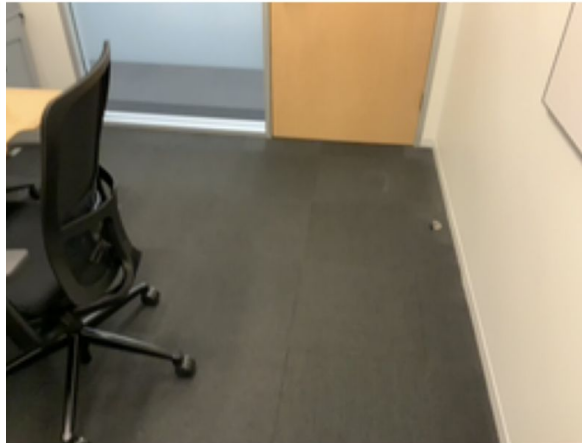
#	query	Object_ present	Map correct	Reasoning
0	ball	FALSE	FALSE	The RGB image does not contain any object that can be identified as a 'ball'. Despite the absence of the object, the similarity map shows significant areas of high similarity (yellow), which is incorrect as there is no 'ball' to highlight.

Edge Case

Semantic Voxel Similarity



RGB Frame 17



17	ball	TRUE	FALSE	A small, light-colored, spherical object resembling a ball is present on the floor near the right wall in the RGB image. However, the similarity map does not distinctly highlight this specific object with high (yellow) similarity. The brightest yellow areas in the map are not co-located with the object.
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VLM-based Classifier - Ground Truth Validation

- **Usage and Rationale:**
 - Done for 2000+ image pairs on the 3 scenes we had.
 - **Pros** : Easy to label a lot of data automatically, gives a base dataset to test other models on.
 - **Cons** : Not a reliable source for GT, but qualitatively, it was fairly accurate on predicting object presence.
- **Other VLMs:**
 - We also tested using a much lighter VLM locally - **Qwen2.5VL** (32B)
 - The results were not bad*,
 - Object Presence ~**87%** similar to Gemini2.5Flash
 - Map correctness ~**52%** similar to Gemini2.5Flash

*Detailed results in Github

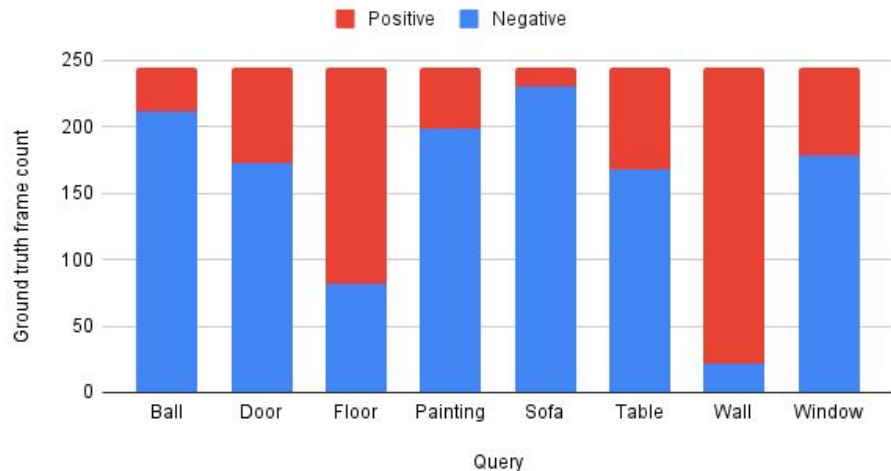
Dataset (Grade C) – ARKitScenes

Scenes: 3 from ARKitScenes; ~80 camera poses per scene (≈ 240 total).

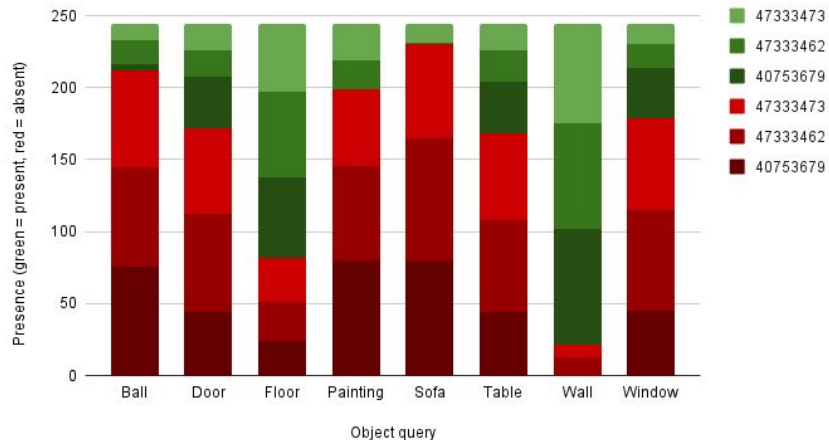
- **Queries/classes (8):** ball, door, floor, painting, sofa, table, wall, window.
- **Total images evaluated per class:** $\sim 240 \rightarrow$ overall $\sim 1,920$ ($\approx 2,000$).
- **Label balance:** ~ 700 positives (object present), $\sim 1,300$ negatives (absent).
- **Artifacts to watch:** small/flat objects (e.g., painting, window), distant 'ball', repeated textures (wall/floor).
- **Deliverables:** per-class counts and example frames.

Statistics on the Ground truth dataset

Ground truth from VLM (all scenes)



Object presence by scene ID



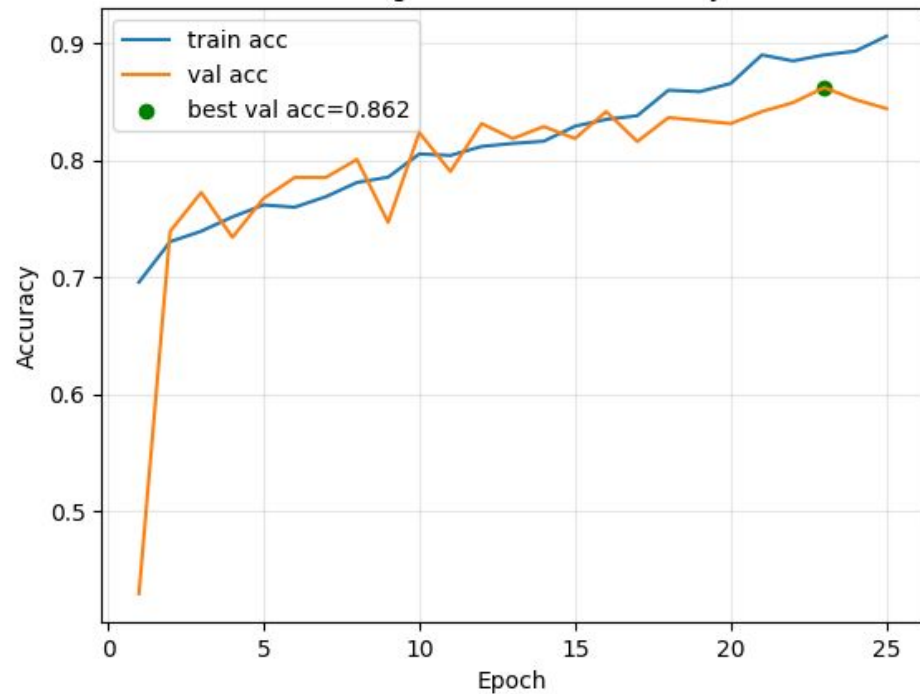
NN-based Classifier (Grade A)

Model: Resnet18 trained from scratch.

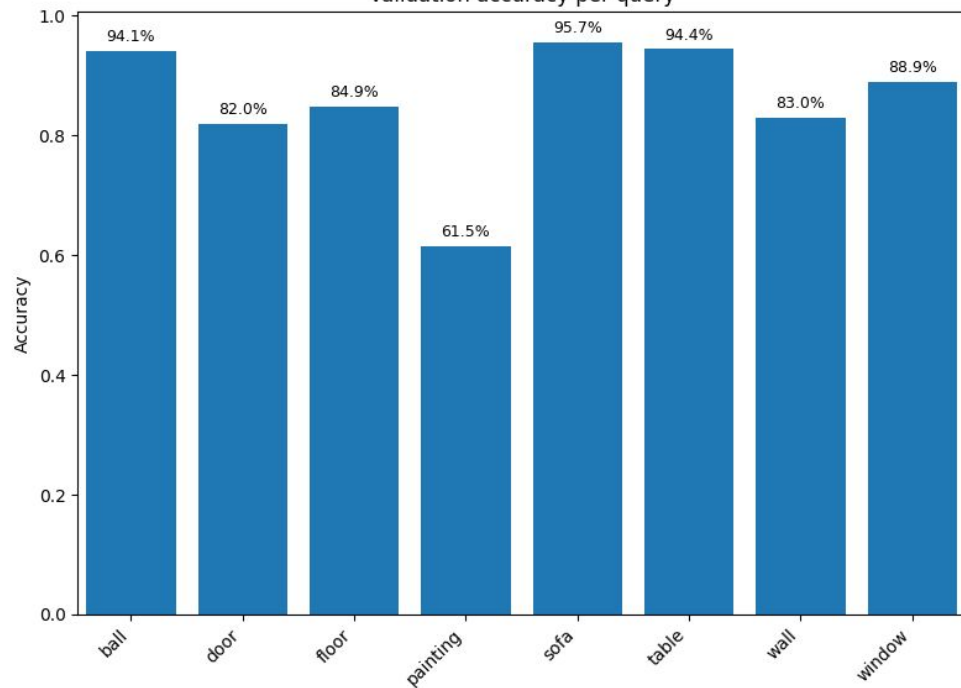
- Training: binary cross-entropy; AdamW.
- 80/20 training/validation split by sequence.
- Validation accuracy = 86.2%; Training accuracy = 90.6%.
- When training from pretrained model overfitting is a big issue.

NN-based Classifier – Results

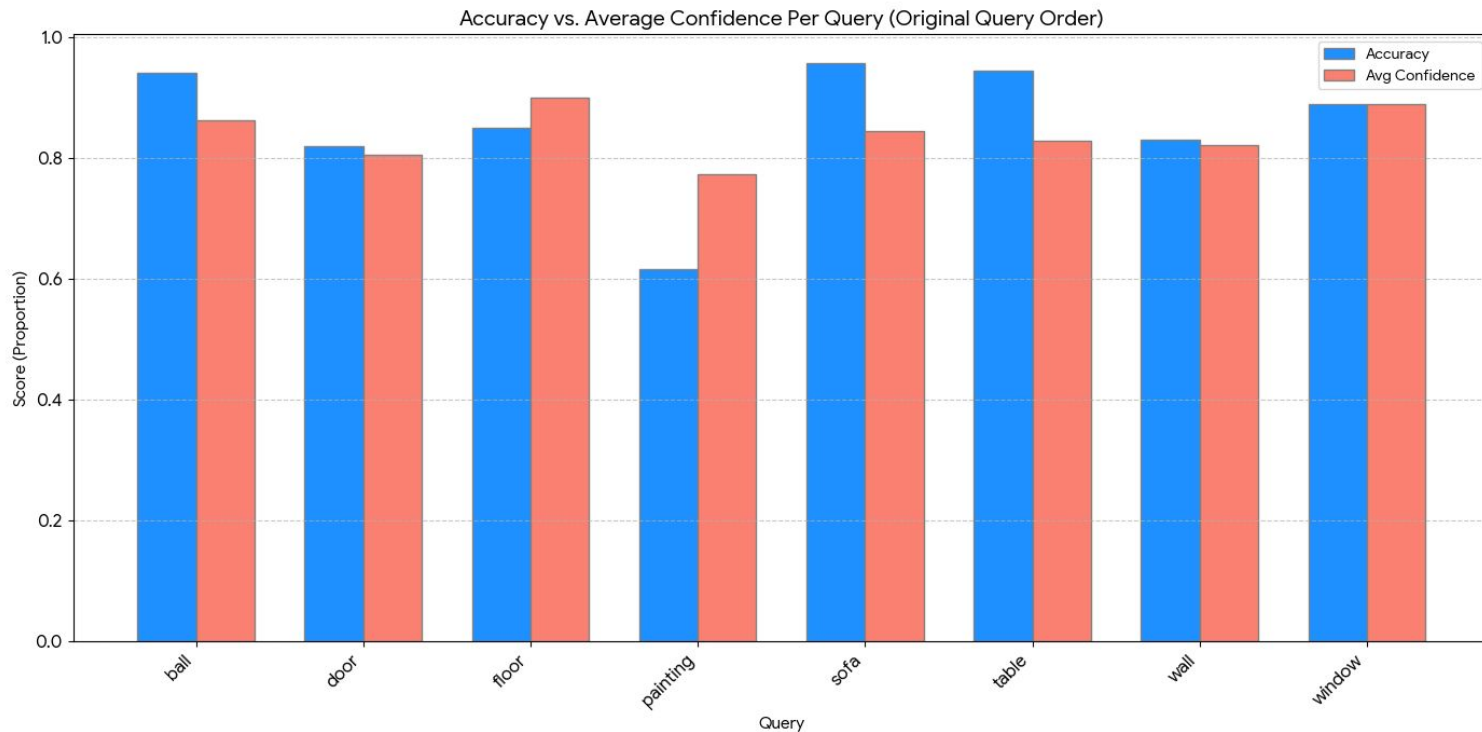
Training vs validation accuracy



Validation accuracy per query



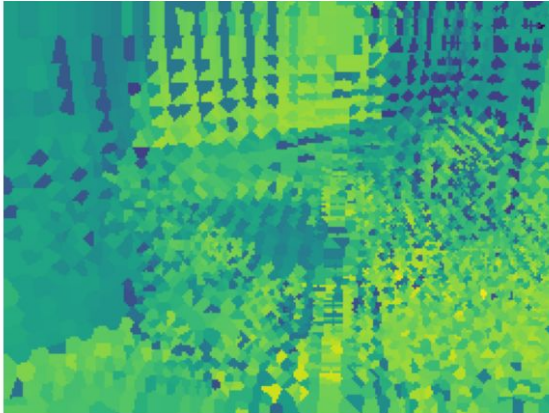
NN-based Classifier – Results



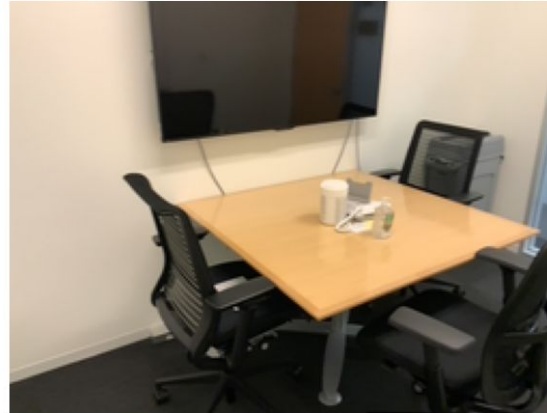
NN-based Classifier – Results

- **True negative** example
- Query: "ball"
- Confidence: 98.1%

Semantic Voxel Similarity



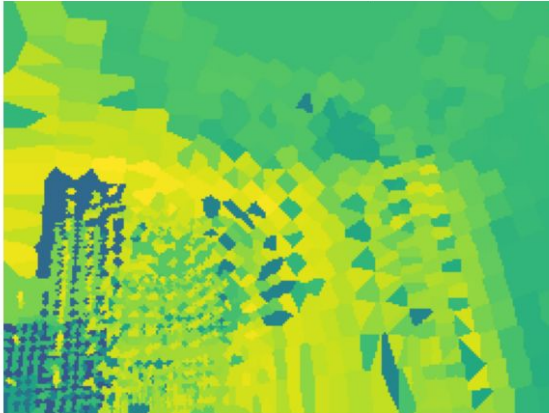
RGB Frame 3



NN-based Classifier – Results

- **True positive** example
- Query: "window"
- Confidence: 96.0%

Semantic Voxel Similarity



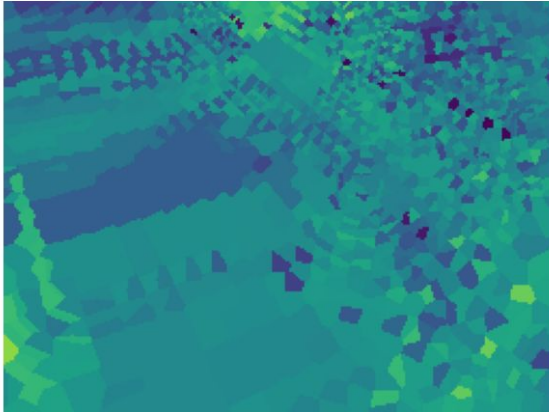
RGB Frame 37



NN-based Classifier – Results

- **False negative** example
- Query: "door"
- Confidence: 99.9%

Semantic Voxel Similarity



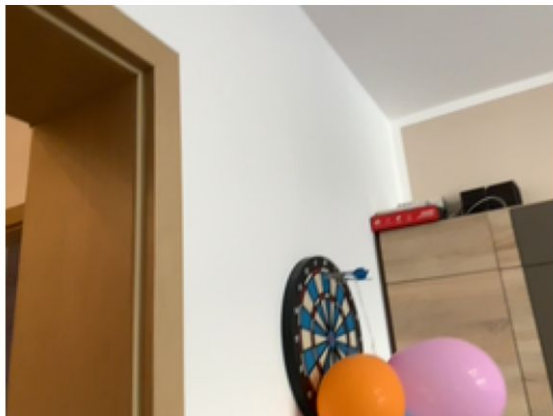
RGB Frame 9



NN-based Classifier – Results

- **False positive** example
- Query: “painting”
- Confidence: 62.9%

RGB Frame 81



Semantic Voxel Similarity



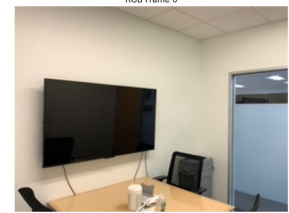
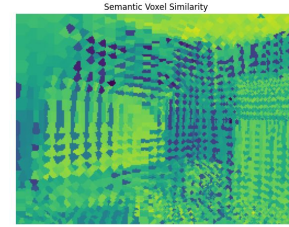


Threshold-based Classifiers

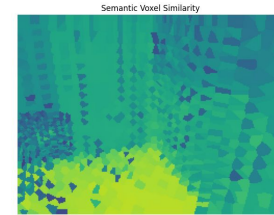
- Predict presence if
 - **Method 1. #pixels above $\tau \geq k$**
 - Method 2. #voxels above $\tau \geq k$

Pixel threshold-based Classifier

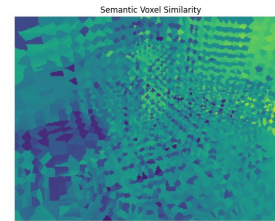
- Predict presence if **#pixels above $\tau \geq k$**
- τ - color
- k - % of image
- **Determining constants :**
 - τ - Optimal 0.8
 - Tested scores from 0.7, 0.8, 0.9
 - k - Optimal 0.1% of the image
 - Tested from (0.1 -> 5% of the img)
- **Results :**
 - Best Accuracy : **74%**



False Positive

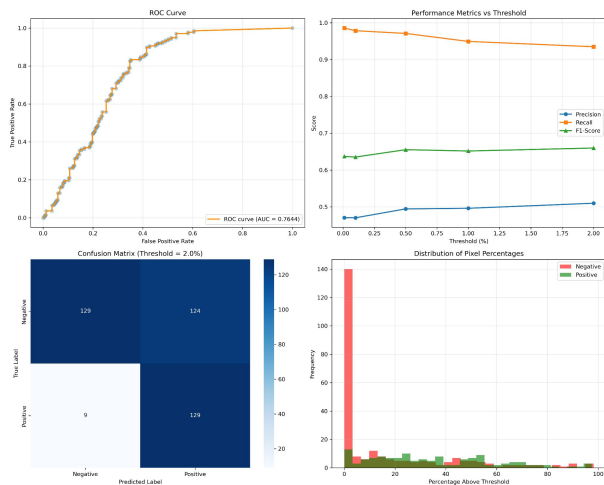


True Positive

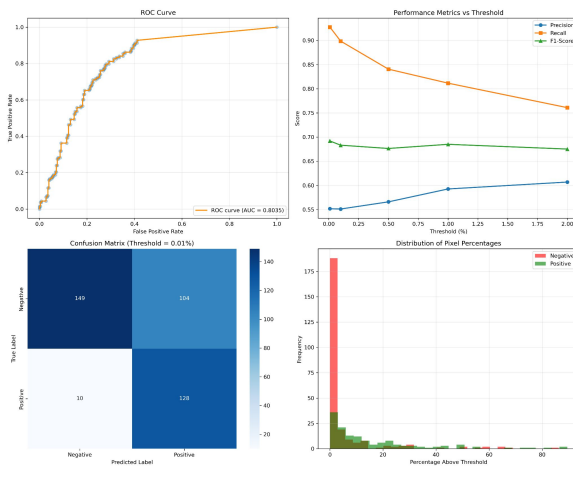


Edge Case : False Positive

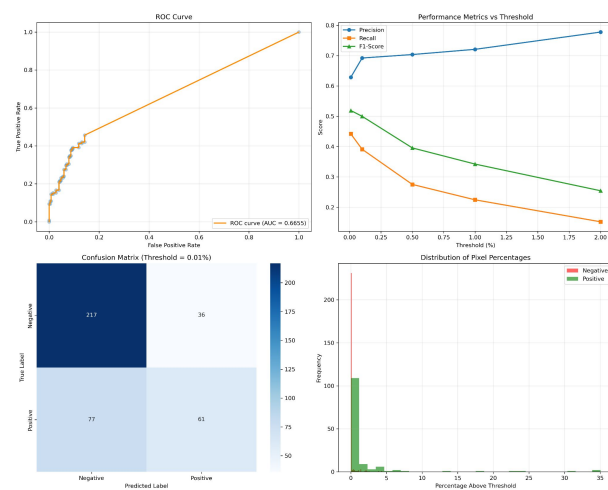
Pixel threshold-based Classifier



~0.7



~0.8



~0.9

Color map from Lab2 :





Voxel threshold-based Classifier

- Predict presence if
 - Method 1. #pixels above $\tau \geq k$
 - **Method 2. #voxels above $\tau \geq k$**

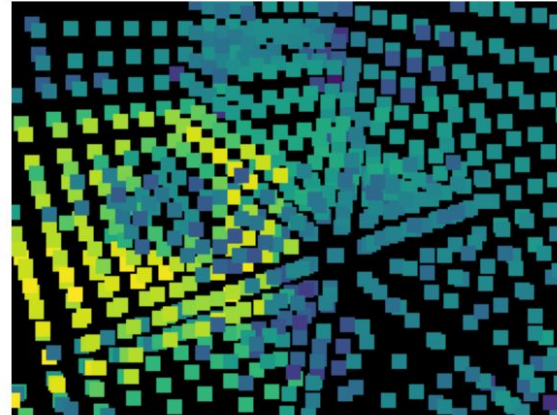
Voxel threshold-based Classifier

- Not based on pixel count in the rendered image but on voxel count in the frame above a certain threshold.
- Based on normalized value (before normalization for the image → consistent).
- Presence if **$\#(\text{voxels} > \text{threshold}) > 10$**

RGB Frame 20

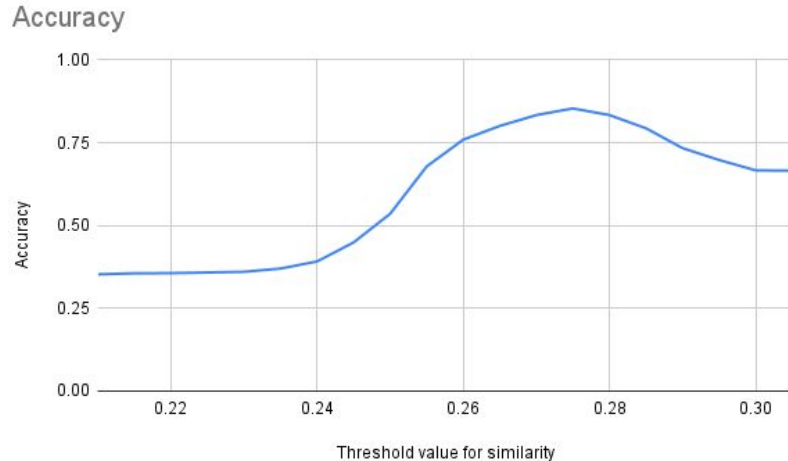


Semantic Voxel Similarity

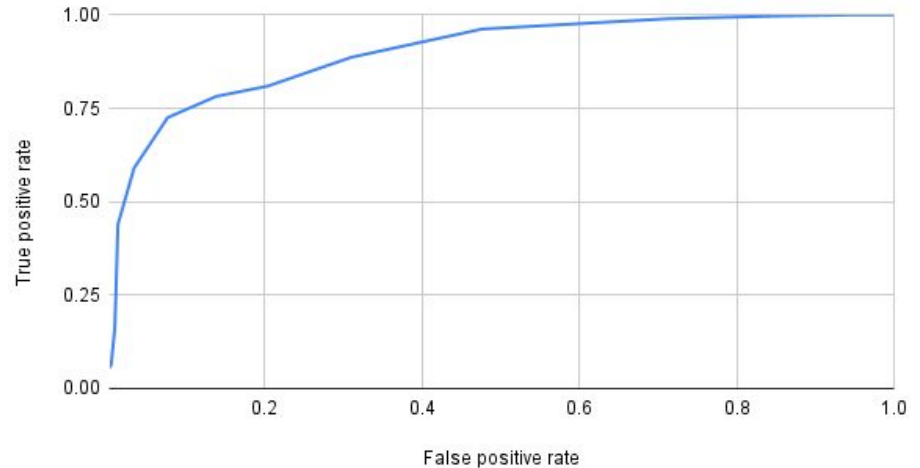


Voxel threshold - analysis

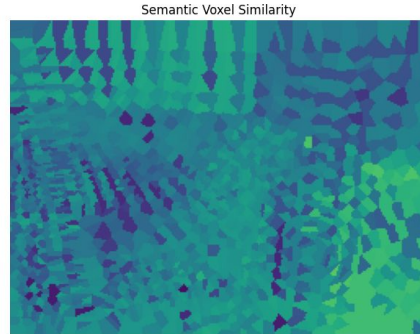
- Determining the optimal threshold value:
 - Optimal threshold = 0.275
 - Optimal accuracy = 85%



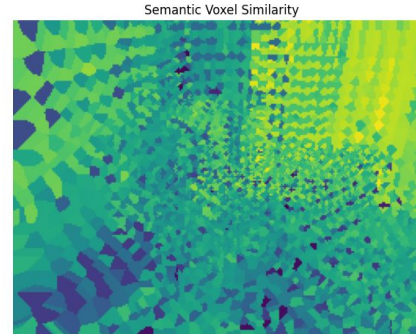
ROC curve for voxel count classifier



Errors for voxel threshold



False positive (door)

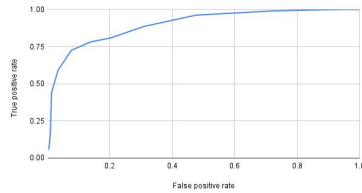


False negative (window)

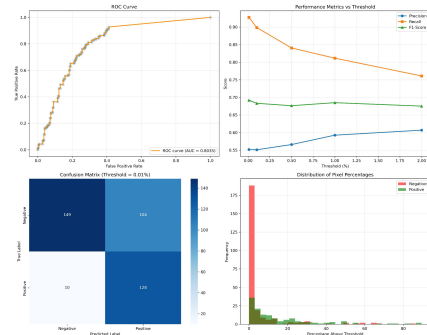
The semantic maps here aren't visually coherent with our intuition because our code normalizes the similarities between 0 and 1 for coloring **before** selecting only the voxels that are in the frame. This is why the right image appears very yellow even though it is a negative and the left one is the inverse (unshown values have biased the coloring, the colors are class-specific).

Results

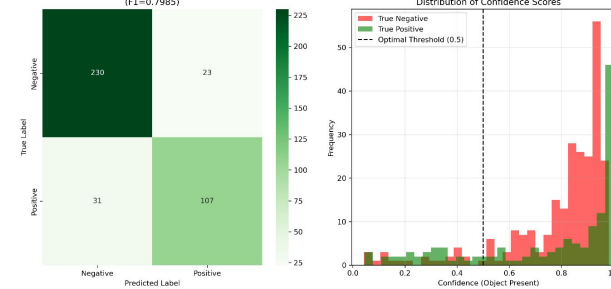
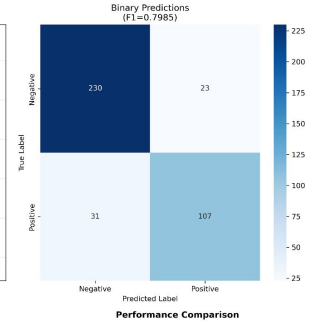
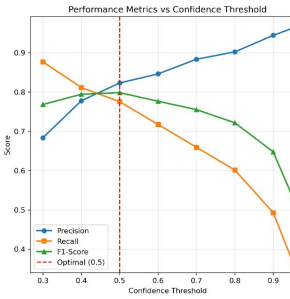
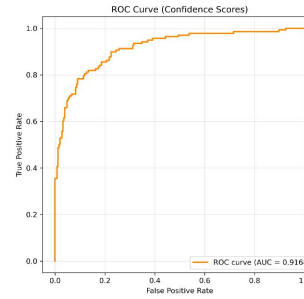
ROC curve for voxel count classifier



Voxel Threshold (85%) - 0.921



Pixel Threshold (74%) - 0.803



NN Based Classifier (86.2%) - 0.917

Metric	Binary Model	Optimal Conf.	Δ
Accuracy	0.8619	0.8619	+0.0000
Precision	0.8231	0.8231	+0.0000
Recall	0.7754	0.7754	+0.0000
F1-Score	0.7985	0.7985	+0.0000



Results

Classifier	Best Validation Accuracy	ROC-AUC
VLM	GT	GT
Pixel Threshold	74	0.803
Voxel Threshold	85	0.921
Resnet	86	0.917

Conclusion

- We can reliably decide object presence from CLIP similarity maps despite scene-wise normalization artifacts that create false “hot” regions when objects are absent.
- **Best model:** ResNet18 presence classifier → 86.2% validation accuracy. Voxel-threshold (absolute similarity, $\tau \approx 0.275$) is close at ~85%; pixel-threshold trails at ~74%
- **Data:** ~2,000 RGB+heatmap pairs from 3 ARKitScenes, 8 classes; label balance ≈ 687 positive / 1,265 negative; gives a meaningful benchmark.
- **Takeaway:** We are train a model to predict object presence, and we think it could perform even better with more data due to overfitting being a problem.

Github: https://github.com/gawtamcr/eai2025_project

Data: [Embodied AI - Results](#)



Thank you!

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